# Athlete Load Monitoring & Injury Risk Prediction

Sports Performance Analytics | XGBoost | SHAP | K-Means

Hugo Ortega-Morales

Analysis Across 74 Elite Dutch Runners throughout 7 Years

# The Problem

## BUSINESS PROBLEM

Injury prevention is essential in elite sport — but coaching staff cannot identify which athletes are at risk without systematic data monitoring. Athletes often feel fine right up until an injury occurs, making intervention impossible without a data-driven signal.

Overtraining is the most common cause. When athletes spike their training load faster than their body can adapt, injury risk increases significantly — yet this risk is invisible to the naked eye.

## AIM

Build a pipeline using 7 years of longitudinal training data to identify high-risk training windows before injuries occur — giving coaching staff actionable signals to intervene proactively.

### Invisible Risk

Athletes feel fine until breakdown — no early warning without data

### Costly Consequences

Injuries impact performance, availability, and medical costs

### Reactive not Proactive

Current monitoring often responds to injury rather than preventing it

# Approach & Key Indicator

## METHODOLOGY

**01 EDA**
Identified total km and wellness metrics as strongest injury signals

**02 ACWR Engineering**
Built Acute to Chronic Workload Ratio — the validated load spike metric

**03 XGBoost Model**
Predicted injury risk from 7-day training windows with time-based split

**04 SHAP**
Explained which features drove model predictions

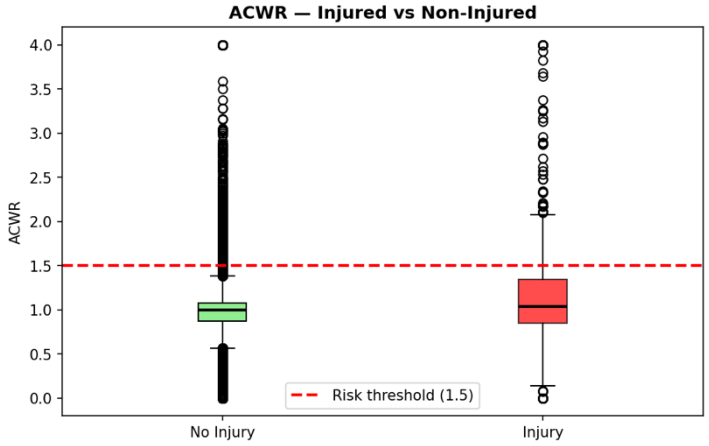
**05 K-Means Clustering**
Grouped athletes into training load profiles for daily monitoring

## THE KEY INDICATOR — ACWR

The Acute to Chronic Workload Ratio (ACWR) measures how hard an athlete is training right now relative to what their body is conditioned to handle.

Acute Load = km run in the last 7 days
Chronic Load = average km over the last 28 days
ACWR = Acute ÷ Chronic

**9x** Higher injury rate when ACWR >= 1.5
(9.04 % vs 1.15% below threshold)



# Predictive Model & Feature Drivers

## MODEL SUMMARY

An XGBoost classifier was trained to predict whether a 7-day training window will result in an injury. The model was trained on data up to 2017 and tested on 2017–2019 to prevent data leakage across overlapping windows.

Class imbalance was handled by penalizing missed injuries 73x more heavily than missed non-injuries.

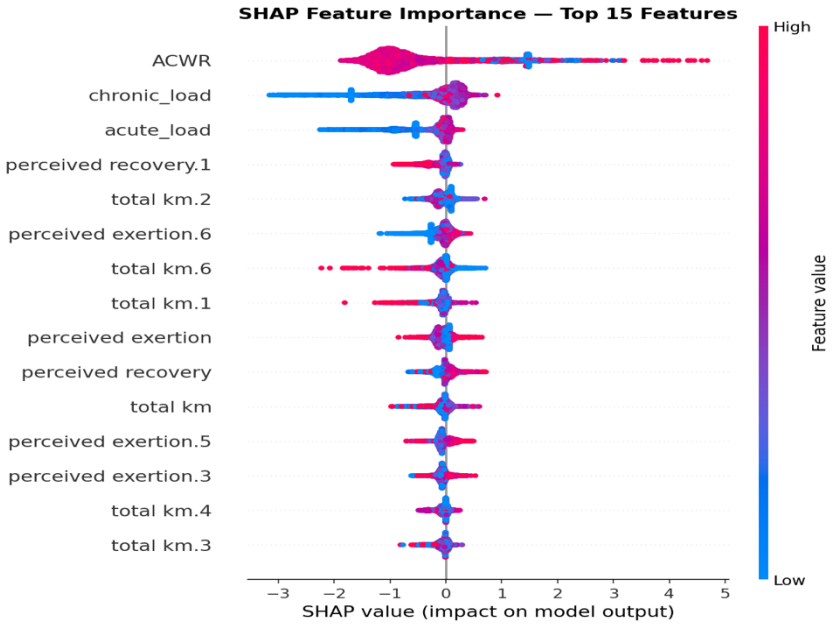
**0.7483**
**AUC-ROC**
vs 0.50 random guess

**27%**
**Injury Recall**
at default threshold

### On Recall

Low recall reflects the 73:1 class imbalance — not a modeling failure. Lowering the decision threshold to 0.30–0.35 would catch more injuries at the cost of more false alarms. This is a coaching staff decision.

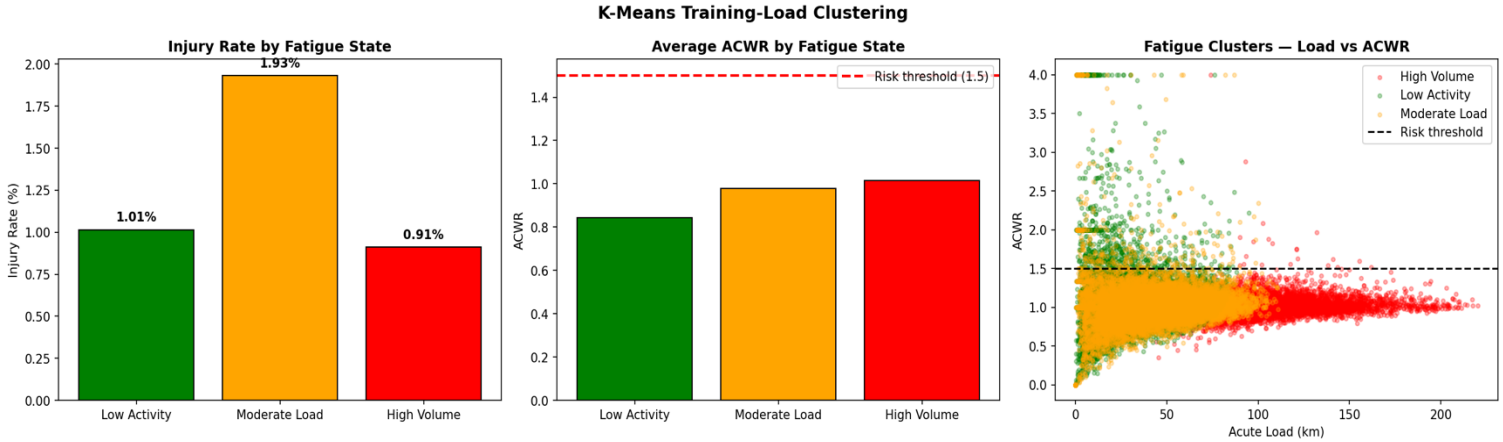
## WHAT DRIVES INJURY RISK — SHAP




**ACWR:** Dominant signal — high values strongly predict injury
**chronic_load & acute_load:** Engineered features rank 2nd and 3rd
**Perceived wellness:** Exertion and recovery across multiple days add signal beyond volume alone

# Findings & Recommendations

## TRAINING LOAD CLUSTERING




**Key Finding:**

High volume athletes rarely exceed ACWR 1.5 despite running 100km+ weeks. Their conditioned baseline protects them. Low activity athletes need only a modest volume increase to spike into the danger zone thus leading to a higher injury rate.

## RECOMMENDATIONS

**01 Flag ACWR ≥ 1.5 immediately**
This single rule captures the highest risk windows — supported by a 9x injury rate increase above that threshold.

**02 Monitor returning & low activity athletes closely**
Their low chronic baseline makes them structurally more vulnerable to load spikes than consistently high-volume athletes.

**03 Track subjective wellness alongside load**
Perceived exertion and recovery across the full 7-day window add meaningful injury signal. High exertion + low recovery warrants a coaching conversation.

**04 Use the model as a supporting tool**
AUC of 0.7483 confirms the model flags elevated risk windows — use it to prompt coach review, not to replace human judgment.

# Next Steps & Limitations

## NEXT STEPS

### Incorporate HRV & Sleep Data

Objective physiological recovery markers would strengthen wellness signals beyond self-report

### Threshold Optimisation

Precision-recall analysis to find the optimal threshold for a specific coaching context

### Extend to Team Sports

Apply pipeline to basketball and football where load monitoring is equally critical

## LIMITATIONS

### Dataset Scope

74 elite distance runners — findings may not transfer directly to team or contact sport athletes

### ACWR Approximation

Chronic load estimated from weekly rolling average rather than true 28-day daily granularity

### Single Load Metric

ACWR built on total km only — intensity zones not incorporated as separate components

### Self-Reported Wellness

Perceived exertion and recovery subject to athlete response bias in competitive environments